

160 kmph

The speed at which it becomes possible for most F1 cars to drive upside down in tunnels, due to high downward forces

Race tracks

In the Monaco Grand Prix, manhole covers had to be welded to avoid dislocation due to the downward force of the cars



There's more to F1 cars than powerful engines. Aerodynamics, gravity and smart design work wonders on the track

F1 for speed

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The modern formula one (F1) race car has a lot in common with an advanced jet fighter. A majority of these common features come from the aerodynamic considerations that go into its design.

To truly understand aerodynamics in F1 cars, we'll have to look at the various forces, aerodynamic and otherwise that act on an F1 race car (or for that matter, any moving body in the atmosphere). One of the largest forces that act on an F1 race car is its thrust - the driving force supplied by the engines, through the wheels, that causes it to move forward (acceleration). In an F1 race car, this force is often several fold greater than regular cars, although the total weight is much lower than a road car.

The second force, known as drag, acts in the opposite direction to its motion (backward) and is due to air resistance. A core function of aerodynamics in F1 race cars is to minimise this drag force. The weight of

the car, caused by gravity is the third force and acts in a downward direction, to keep the car to the ground. A fourth, very important force, is the aerodynamically created downward force, which acts more or less in the same direction as the weight of the car, to keep it pressed downward. In modern F1 cars, the down force can reach magnitudes as high as 3.5 times the weight of the car.

While the driver can vary thrust by changing gears and revving the engine, the weight is constant. The drag and downward force are largely a function of the speed of the car (and hence not directly under the driver's control).

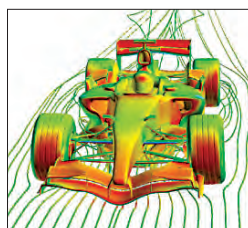
Function of aerodynamics

In the 1960s, with stagnation in internal combustion engine technologies (all path breaking developments had already been made, with further developments leading to only minor refinements), new ways had to be explored to improve the performance of a race car.

One of the first steps

in this direction was the realisation that the weights of the cars could be further dropped (allowing them to become lighter and hence faster), while the function that weight served, of keeping the car pressed to the road during curves and slopes, could be passed on to aerodynamic forces. The result of this line of thought was to fit wings onto F1 race cars. Though similar in many ways to aircraft wings (in their cross sectional geometries), the wings in F1 race cars are mounted upside down, (in a manner of speaking) to create down force rather than lift (the upward force that helps an aircraft fly).

Alongside, the overall geometries of an aircraft were also altered to reduce drag, so that the total force causing it to accelerate could be increased. The result was rapid and explosive development in these sectors often leading to catastrophic disasters, causing the racing regulatory authorities to follow up with strict regulations, in the interest of safety. The 1970 racing season witnessed a new set of regulations, on the size and location of wings in an F1 race car. Ever since, the aerodynamics have played a large, often pivotal role in F1 race car



Airflow over a Renault formula 1 race car, generated using the software Fluent 6